

JAMES MITCHELL

Research Engineer

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PROFESSIONAL SUMMARY

Experienced Research Engineer with 12 years in designing and implementing advanced data systems, machine learning infrastructure, and distributed computing solutions. Proven track record of translating complex research concepts into production-grade systems while leading cross-functional teams. Passionate about solving challenging technical problems at scale and mentoring junior engineers to develop their expertise.

WORK EXPERIENCE

Research Engineer - Staff Level

DataStream Analytics Inc

Remote / Edinburgh, United Kingdom

<https://datastreamanalytics.io>

January 2021 - Present

Architected and deployed machine learning pipeline infrastructure handling 50+ terabytes of daily data, supporting real-time analytics for enterprise clients across financial services and telecommunications sectors. Led a team of 4 engineers in designing microservices-based architecture using Kubernetes and Apache Kafka for fault-tolerant event processing. Implemented advanced data transformation workflows using HuggingFace Transformers and LangChain for natural language processing tasks, achieving 35% improvement in model inference latency. Established best practices for reproducible research including containerization, versioning, and automated testing frameworks that reduced deployment time by 60%.

Senior Research Engineer

Skyscanner

Edinburgh, United Kingdom

<https://www.skyscanner.net>

November 2019 - March 2021

Developed core recommendation engine using Apache Spark and Python that increased booking conversion rates by 18% across mobile and web platforms serving 100 million monthly active users. Designed and implemented feature engineering pipeline with dbt for data transformation, ensuring data quality and consistency across 200+ models in production analytics environment. Pioneered internal research initiative on graph neural networks for travel route optimization, publishing findings in 3 peer-reviewed conferences and securing 2 million GBP in follow-up funding. Mentored 6 junior engineers on machine learning best practices, code review processes, and system design principles, with 4 subsequently promoted to senior roles.

Research Engineer

Heriot-Watt University

Edinburgh, United Kingdom

<https://www.hw.ac.uk>

July 2012 - November 2019

Conducted research in distributed systems and cloud computing, developing novel algorithms for resource allocation in heterogeneous environments with publications in IEEE Transactions and ACM conferences.

Implemented continuous integration pipelines using GitHub Actions for automated testing and deployment of research code, reducing manual testing overhead by 75% and improving code reliability. Designed and delivered 8 courses on software engineering and systems architecture to postgraduate students, receiving consistently positive feedback with average ratings of 4.7/5.0. Collaborated with industry partners including Microsoft and Amazon Web Services on applied research projects, securing 850,000 GBP in external research grants and supervision of 12 PhD students.

EDUCATION

PhD Computer Science

University of Edinburgh

Edinburgh, United Kingdom

2009 - 2012

BSc Honours Computer Science

University of Manchester

Manchester, United Kingdom

2005 - 2009

TECHNICAL SKILLS

Programming Languages: Python, Java, Scala, SQL, Go, C++

Big Data Technologies: Apache Spark, Apache Kafka, Hadoop, Apache Flink, Databricks

Cloud Platforms: Amazon Web Services, Google Cloud Platform, Microsoft Azure

Machine Learning: TensorFlow, PyTorch, Scikit-learn, HuggingFace Transformers, MLflow

Data Engineering: Airflow, dbt, Datadog, ELK Stack, Prometheus

Distributed Systems: Kubernetes, Docker, Microservices, Event-driven Architecture

Databases: PostgreSQL, MongoDB, Cassandra, DynamoDB, Redis

Version Control: Git, GitLab, Bitbucket

Additional: System Design, Research Methodology, Technical Leadership, Performance Optimization